

Multiple Choice

1. **Answer:** A

Options: A) Through vertical gene transfer from parent to offspring; B) Through random assortment of genes during meiosis; C) Through peptide bond formation errors during translation; D) Through genetic mutation only; E) Through transcriptional errors in RNA processing

Note: The correct answer is based on the understanding that vertical gene transfer allows genetic information, including diverse protein-coding genes, to be passed from parent organisms to their offspring, thereby enabling the expression of a wide variety of proteins within a population over generations.

2. **Answer:** A

Options: A) The gills of a fish and the lungs of a human; B) The leaves of an oak tree and the spines of a cactus; C) The eyes of a human and the eyes of a squid; D) The roots of a tree and the legs of an animal; E) The presence of a notochord in all chordate embryos

Note: The gills of a fish and the lungs of a human exemplify convergent evolution because they are both adaptations for breathing in different environments-water and air, respectively-demonstrating how similar functional solutions can arise independently in distinct evolutionary lineages.

3. **Answer:** A

Options: A) All organisms reproduce.; B) All organisms are capable of evolving.; C) All organisms require energy.; D) All organisms have DNA.; E) All organisms are composed of cells.

Note: The fact that all organisms reproduce demonstrates a fundamental biological principle shared across diverse life forms, indicating a common evolutionary ancestry and genetic continuity among all living entities on Earth.

4. **Answer:** A

Options: A) Oxygen is used to maintain the temperature of cells.; B) Oxygen is used by cells to produce alcohol.; C) Oxygen is used to produce carbon dioxide in cells.; D) Oxygen is solely responsible for stimulating brain activity.; E) Oxygen is used to synthesize proteins in cells.

Note: Oxygen plays a crucial role in cellular respiration, a process that generates the energy needed for maintaining cellular functions and regulating temperature in the body.

5. **Answer:** A

Options: A) Animals use the temperature changes as cues.; B) Animals use the color of the surroundings as cues.; C) Animals use the noises from the environment as cues.; D) Animals navigate by following other migrating animal species exclusively.; E) Animals

rely solely on the memory of previous migration paths.

Note: Animals use temperature changes as cues for migration because these variations often signal seasonal changes and help them determine the optimal timing for their migratory journey to ensure survival and reproductive success.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The offspring of a person with O type blood (genotype ii) and a person with AB type blood (genotype IAIB) can only inherit the A allele from the AB parent or the B allele from the AB parent, resulting in a 50% chance of being type A (IAi) and a 50% chance of being type B (IBi).

7. **Answer:** False

Options: A) True; B) False

Note: Guttation is primarily caused by root pressure and the plant's physiological processes rather than external weather conditions, as it occurs when water is pushed up from the roots through special structures called hydathodes, typically during the night when transpiration is low.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the enzyme's constant activity across pH values suggests that factors other than ionizing groups, such as the enzyme's overall structure and non-ionic interactions, are also crucial for its catalytic function.

9. **Answer:** False

Options: A) True; B) False

Note: The probability of drawing the ace of diamonds from a standard deck of cards is $1/52$, not $1/53$, because a standard deck contains 52 cards.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because a higher pH value indicates a more alkaline condition, and if Lake B has a higher pH than Lake A, it should indeed be more alkaline, contradicting the given assertion.

Long Form Response

11. **Answer:** To determine whether the newly discovered bioluminescent worm is the source of light production rather than a contaminating bacteria, I would employ two primary methods. First, I would place the light-producing substance in complete growth media to observe if the light emission persists in the absence of other potential light sources. This would help assess whether the worm itself has the intrinsic capability

for bioluminescence. Second, I would conduct a physical examination of the worm under a microscope to inspect for any bacterial contamination that could be responsible for the luminescence. By integrating these approaches, I can gather evidence to confirm the origin of the light and distinguish between the worm and any possible contaminants. *Note:* correct because it outlines a systematic approach that includes isolating the light-producing substance to determine if the worm possesses intrinsic bioluminescence, thereby distinguishing it from potential contamination by bacteria.

12. **Answer:** Whales have developed several adaptations that enable them to thrive in cold water environments, particularly through their thermoregulation and insulation mechanisms. One of the key adaptations is their ability to maintain a nearly constant internal body temperature, which is critical for metabolic functions and overall health. To achieve this, whales possess a significant layer of subcutaneous fat, known as blubber, which acts as an insulating barrier against the cold water. This blubber not only helps to retain body heat but also provides energy reserves during times of fasting, such as during migration or breeding. Additionally, the thick skin of whales further aids in reducing heat loss, allowing them to effectively adapt to their frigid habitats. *Note:* correct because it accurately identifies the critical role of blubber in thermoregulation and insulation, which allows whales to maintain a stable internal body temperature necessary for survival in cold water environments.